FRACMATH: numerical damage update

The modified von Mises equivalent-strain function used in the diagram is

$$\widetilde{\varepsilon}_{\mathrm{mvM}}(I_1, J_2) = \frac{k-1}{2k(1-2\nu)} I_1 + \frac{1}{2k}\sqrt{\left(\frac{k-1}{1-2\nu}\right)^2 I_1^2 + \frac{12k}{(1+\nu)^2} J_2},$$
$$k = f_c/f_t.$$

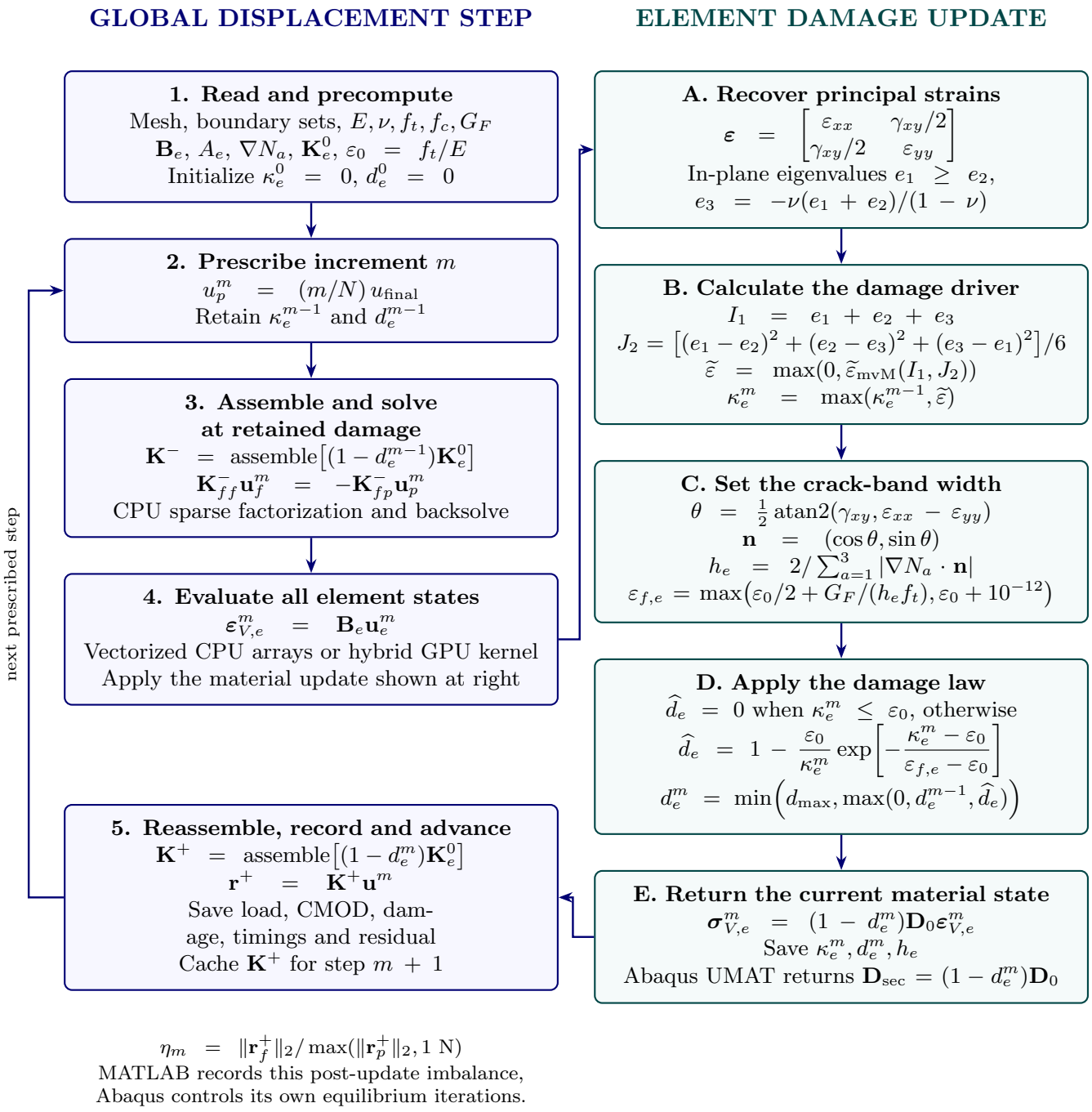


Figure 1: 2D solution procedure and element damage update. Here N counts increments, $\mathbf{u}$ is the nodal displacement vector, and u_{final} is the final imposed displacement. The matrix $\mathbf{B}_e$ maps element displacements to engineering strain; $\mathbf{K}_e^0$ is undamaged element stiffness. Superscripts $-$ and $+$ identify stiffness before and after the damage update. The internal force vector is $\mathbf{r}^+ = \mathbf{K}^+\mathbf{u}$; subscripts f and p identify free and prescribed degrees of freedom. The relative imbalance η_m compares free-force and reaction magnitudes with a 1 N lower bound. The Euclidean norm $\|\mathbf{v}\|_2 = \sqrt{\sum_i v_i^2}$ gives vector magnitude. The angle θ defines the crack normal, and atan2 is the two-argument inverse tangent. Trial damage is $\widehat{d}_e$; the UMAT returns secant stiffness $\mathbf{D}_{\mathrm{sec}}$. Code-name mappings are given in `NOTATION.md`.